

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

Roll No. :52	Name:ROSHAN YADAV
Class :SYIT	Batch : 1
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Queue Application:

- a. Write a program to: a. Implement a queue using an array.

code

```
from array import array
```

```
queue = array('i') # Integer array
```

```
# Enqueue
```

```
queue.append(10)
```

```
queue.append(20)
```

```
queue.append(30)
```

```
print("Queue:", queue)
```

```
# Dequeue
```

```
print("Deleted:", queue[0])
```

```
queue.pop(0)
```

```
print("Queue after Dequeue:", queue)
```

output

```
Queue: array('i', [10, 20, 30])
Deleted: 10
Queue after Dequeue: array('i', [20, 30])
```

- b. Simulate a simple queuing system (e.g., customer service queue).

code

```
from array import array
```

```
# Create Queue
customers = array('i')

# Add Customers
customers.append(101)
customers.append(102)
customers.append(103)

print("Customers in Queue:", customers)

# Serve First Customer
print("Served Customer:", customers[0])
customers.pop(0)

print("Queue After Serving:", customers)
```

output

```
===== RESTART: C:/Users/IT-43/Desktop/roshn:
Customers in Queue: array('i', [101, 102, 103])
Served Customer: 101
Queue After Serving: array('i', [102, 103])
```

